Ethical Implications of AI on Database Administrators

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As someone who is currently majoring in cybersecurity hoping to pursue a career in database administration, it's a known thing that it's important to keep data safe and making sure its stored and managed in the correct ways (U.S. Bureau of Labor Statistics, 2023). As technology continues to grow, especially artificial intelligence (AI), this job will change drastically. When saying it will change drastically this means AI will not eliminate the job but instead change the way things are done within the job. AI will more than likely change all the repetitive daily task that include setting up databases, performing and monitoring backs ups and recoveries, and managing user access. This means all database administrators (DBAs) will more than likely have to work differently and change and adapt the way things are done.

AI Impact

Artificial intelligence is already changing many jobs within the cybersecurity field. AI can do many task that people are used to manually doing. In database administration, it can help detect problems faster than humans can; it is able to make decisions faster than the human brain can. This is something that most may thing is making the job as a DBA easier, which in most ways is but it also comes with problems. If people are relying on AI to make too many decisions, it could make a lot of simple mistakes. A huge problem is privacy; artificial intelligence systems are able to look through a user's personal information that shouldn't be accessed without permission from the user. Another issue of artificial intelligence in database administration field is data being misused, especially if it's not overlooked by a human being.

Ethical Analysis

One ethical method includes Penn State, which is a process designed to help people navigate through tough decisions. This method gathers the facts, determines the ethical issue,

considers ethical principles and responsibilities, identifies those who are impacted, and finally explores the possible solutions and actions. Penn State may ask questions that include: What could go wrong? Who will be directly impacted by our decision? (Penn State University, n.d.). Using a method like this allows you to see that artificial intelligence in database administration effects more than just its workers but the personal data that is being stored. Looking at this from a Christian Worldview, we are taught to treat others how we except to be treated back; verse Micah 6:8 says that we should always "act justly, to love mercy, and to walk humbly" (Bible Gateway, n.d.). This verse is something that tells us it is required, it is a calling and we shouldn't let artificial intelligence affect others even if it makes our lives easier. Protecting user's private information is very important, we all have value. The Association for Computing Machinery (ACM) and Institute of Electrical and Electronics Engineers (IEEE) both have morals that align with protecting human well-being and public safety (ACM, n.d.; IEEE, n.d.). Both of these groups say that people working with technology should respect people's privacy and avoid all harm. These polices and programs imply that it is important to use artificial carefully and follow ethical rules to protect communities.

Comparison

When comparing Christian morals with these different policies and programs there are similarities and differences. It is said from all of them that we should avoid harm and provide fairness to all because we all have value. They all also imply that people should take responsibility for actions made. A difference that can be seen is Christian Worldview adds a much more deeper meaning behind it. It gives a command that every person is made in the image of God. We should treat people with love and kindness always. While people find these policies and programs helpful to understand, the Christian Worldview goes beyond rules and regulations.

Both viewpoints are extremely helpful when looking at these kinds of things, but the Christian view just takes an approach that allows us to know that ethics is about our values not just policies and programs.

Conclusion

When maneuvering through the ethics of artificial intelligence, the best solution is using AI as a tool and keeping people in control. We know that AI is more than capable of handling task all within database administration, but it's important to review its work every time. This approach allows all data to be safe, stored, and managed in the correct way. All companies should start teaching workers not only in just technology but every field about ethical viewpoints. An idea that could be tested and would fail because it's not ethical would include letting AI run databases without any human help. This is extremely risky and would hurt many people. In the end, it's important to takeaway that AI can be used in both a good and fair way if looked at ethically.

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